Financial Reporting - Claims Data mart

The Claims Data Mart is a structured data warehouse that consolidates claims financial transactions, reserves, payments, and recoveries for financial reporting. This fact table query extracts key claims financial data across different time periods (Month-to-Date (MTD), Quarter-to-Date (QTD), and Year-to-Date (YTD)) for analytics

This fact table query captures financial data at multiple granularity levels, linking claims to their respective exposures, policies, regions, and financial transactions. It enables insurers to generate detailed claims financial reports for reserve management, payment tracking, and recovery analysis.

Note : - Suggesting to visit this page, as it will provide basic terminology and claims related details.

Fact Table Overview

The fact table query extracts key claims financial data at multiple levels, linking claims to their respective exposures, policies, coverages, regions, and financial transactions. This enables insurers to:

- Track claim reserves over time.
- Analyse indemnity and medical payments.
- Monitor subrogation and salvage recoveries.

Github link - Claims data mart

Source Tables

The fact table extracts and aggregates data from multiple transactional sources:

```
FROM
  dd_month_crv.dd_month_crv month
  LEFT JOIN df_loss_transactionlineitem_crv.df_loss_transactionlineitem_crv
transline
  ON month.bkey=cast(to_char(transline.DN_TransactionID_BookingDate,
'yyyyMM') as bigint)
  Left JOIN dd_loss_transaction_crv.dd_loss_transaction_crv trans
  on trans.skey = transline.FK_TransactionID_SKEY
  left JOIN wrk_pc_cc_integration.wrk_pc_cc_integration wrk_integration
  on transline.FK_TransactionID_ExposureID_SKEY =
wrk_integration.fk_exposureid_skey
  LEFT JOIN df_loss_claimstatusmonthend_crv.df_loss_claimstatusmonthend_crv
claimstatus
  on transline.FK_TransactionID_ClaimID_BKEY=claimstatus.FK_ClaimID_BKEY
```

```
  LEFT JOIN
df_loss_exposurestatusmonthend_crv.df_loss_exposurestatusmonthend_crv
exposurestatus
  on transline.FK_TransactionID_ClaimID_BKEY=exposurestatus.FK_ClaimID_BKEY
and transline.FK_TransactionID_ExposureID_BKEY =
exposurestatus.FK_ExposureID_BKEY
  where month.yearid between (SELECT min(extract('year' from
transline.DN_TransactionID_BookingDate)) AS min_year FROM
df_loss_transactionlineitem_crv.df_loss_transactionlineitem_crv transline)
  and (SELECT max(extract('year' from
transline.DN_TransactionID_BookingDate)) AS max_year FROM
df_loss_transactionlineitem_crv.df_loss_transactionlineitem_crv transline)
```

Source Table	Description
`dd_month_crv`	Provides calendar-based reference data for months, quarters, and years.
`df_loss_transactionlineitem_crv`	Stores detailed transaction line items for claim payments, reserves, and recoveries.
`df_loss_transaction_crv`	Transaction details for claim
`wrk_pc_cc_integration`	Contains claim, policy, and exposure mappings between Guidewire PolicyCenter (PC) and ClaimCenter (CC).
`df_loss_claimstatusmonthend_crv`	Tracks claim status as of the end of each month.
`df_loss_exposurestatusmonthend_crv`	Tracks exposure-level status as of the end of each month.

Fact Table Schema Explanation

Primary keys, Foreign keys, De-norm columns:

Column Name	Description
`bkey`	Unique key combining claim and time period.
`skey`	Hashed unique key for record identification.
`fk_snapshotperiod_bkey`	Accounting period Month
`quarter_id`	Accounting period Quarter
`year_id`	Accounting period Year
`dn_productcode`	Line of Business

`fk_datamartdimkeys_skey`	Primary key linking to dimension tables. This is mainly used to do the windowing sum.
`fk_uwcompany_skey`	Underwriting Company
`fk_branchid_skey`	Policy Period ID which is associated with the policy
`fk_region_bkey`	Region where the claim is filed/happened.
`fk_coverageid_skey`	Coverage linked to the claim.
`fk_claimid_skey`	Claim associated
`fk_claimantdenormid_skey`	Claimant associated with this claim
`fk_exposureid_skey`	Exposure associated with the claim.
`fk_producercodeofrecordid_skey`	Producer responsible for policy
`fk_producercodeofrecordid_organizationid_skey`	Producer's Organization
`fk_pnicontactdenorm_skey`	Primary Named Insured
`fk_accountid_skey`	Account to which policy is assoicated
`TC_Currency`	Currency for financial transactions.
`CF_claim_MonthEndStatus`	Claim status at month-end.
`CF_exposure_MonthEndStatus`	Exposure status at month-end

`fk_datamartdimkeys_skey` is coming from the `wrk_pc_cc_integration` dataset, which is just a MD5 of all dimension's primary key. This is used mainly in windowing function for providing the partitions in ITD, QTD, YTD calculation.

```
md5(
      coalesce(pp.fk_uwcompany_skey, 'NONE')
       || '-' || coalesce(pp.skey, 'NONE')
       || '-' || coalesce(pppl.fk_region_bkey, 'NONE')
       || '-' || coalesce(e.fk_coverageid_skey, 'NONE')
       || '-' || coalesce(c.skey, 'NONE')
       || '-' || coalesce(coalesce(e.fk_claimantdenormid_skey,
c.fk_claimantdenormid_skey), 'NONE')
       || '-' || coalesce(e.skey, 'NONE')
       || '-' || coalesce(pp.fk_producercodeofrecordid_skey, 'NONE')
       || '-' || coalesce(pp.fk_producercodeofrecordid_organizationid_skey,
'NONE')
       || '-' || coalesce(pp.fk_pnicontactdenorm_skey, 'NONE')
       || '-' || coalesce(pp.fk_policyid_accountid_skey, 'NONE')
    ) as fk_datamartdimkeys_skey
```

Calculation Columns:

These fields track cumulative financial data from the start of the claim until the reporting period (ITD):

Column Name	Description
CF_itd_indemnityreservechangeamount	Cumulative indemnity reserve changes.
CF_itd_medicalreservechangeamount	Cumulative medical reserve changes.
CF_itd_expenselaereservechangeamount	Cumulative Loss Adjustment Expense (LAE) reserve changes.
CF_itd_expenseulaereservechangeamount	Cumulative Unallocated Loss Adjustment Expense (ULAE) reserve changes.
CF_itd_salvagereservechangeamount	Cumulative salvage reserve changes.
CF_itd_subroreservechangeamount	Cumulative subrogation reserve changes.

The Below fields tracks the financial data for reserves, payments, recoveries at different timelines. **The below table only defines the columns for monthly calculation but ,these will be calculated at each month, quarter and year timeline level.**

The following table consolidates all Month-To-Date (MTD) financial aggregations used in the query. These metrics track financial transactions within the current month, ensuring accurate reporting of reserves, payments, and recoveries.

Column Name	SQL Alias	Description
Payments		
MTD Indemnity Payments	CF_mtd_paymentindemnityamount	Total indemnity payments made this month.
MTD Medical Payments	CF_mtd_paymentmedicalamount	Total medical payments made this month.
MTD Expense LAE Payments	CF_mtd_paymentexpenselaeamount	Total Loss Adjustment Expense (LAE) payments made this month.
MTD Expense ULAE Payments	CF_mtd_paymentexpenseulaeamount	Total Unallocated Loss Adjustment Expense (ULAE) payments made this month.
Recovery		

MTD Recovery Deductible Amount	`CF_mtd_recoverydeductibleamount`	Total deductible recoveries received this month.
MTD Recovery Salvage Amount	`CF_mtd_recoverysalvageamount`	Total salvage recoveries received this month.
MTD Recovery Subrogation Amount	`CF_mtd_recoverysubroamount`	Total subrogation recoveries received this month.
MTD Recovery Indemnity Amount	`CF_mtd_recoveryindemnityamount`	Total indemnity recoveries received this month.
MTD Recovery Expense Amount	`CF_mtd_recoveryexpenseamount`	Total expense recoveries received this month.
MTD Recovery Expense Salvage Amount	`CF_mtd_recoveryexpensesalvageamount`	Total salvage-related expenses recovered this month.
MTD Recovery Expense Subrogation Amount	`CF_mtd_recoveryexpensesubroamount`	Total subrogation-related expenses recovered this month.
Reserve		
MTD Indemnity Reserve Change Amount	`CF_mtd_indemnityreservechangeamount`	Total indemnity reserve adjustments (increase/decrease) this month.
MTD Medical Reserve Change Amount	`CF_mtd_medicalreservechangeamount`	Total medical reserve adjustments (increase/decrease) this month.
MTD Expense LAE Reserve Change Amount	`CF_mtd_expenselaereservechangeamount`	Total Loss Adjustment Expenses (LAE) reserve adjustments (increase/decrease) this month.

MTD Expense ULAE Reserve Change Amount	`CF_mtd_expenseulaereservechangeamount`	Total Unallocated Loss Adjustment Expenses (ULAE) reserve adjustments (increase/decrease) this month.
MTD Salvage Reserve Change Amount	`CF_mtd_salvagereservechangeamount`	Total changes in salvage reserves recorded this month.
MTD Subrogation Reserve Change Amount	`CF_mtd_subroreservechangeamount`	Total changes in subrogation reserves recorded this month.

Summary of MTD Aggregations

- All metrics represent financial activity for the current month only.
- Payments track actual disbursements made for indemnity, medical, and expenses.
- Recoveries track incoming funds from subrogation, deductible payments, and salvage.
- Reserve Changes track adjustments in estimated reserves for future payments and the eroding of reserves while making the payments.

As these calculation will happen at quarter and year timeline line as well.

Understanding Aggregation Behavior in the Claim DataMart Fact Table

```
group by month.bkey,
month.quarterid,
month.yearid,
wrk_integration.dn_policyid_productcode,
wrk_integration.fk_datamartdimkeys_skey,
wrk_integration.fk_uwcompany_skey,
wrk_integration.fk_branchid_skey,
wrk_integration.fk_region_bkey,
wrk_integration.fk_coverageid_skey,
wrk_integration.fk_claimid_skey,
wrk_integration.fk_claimantdenormid_skey,
wrk_integration.fk_exposureid_skey,
wrk_integration.fk_producercodeofrecordid_skey,
wrk_integration.fk_producercodeofrecordid_organizationid_skey,
wrk_integration.fk_pnicontactdenorm_skey,
wrk_integration.fk_accountid_skey,
trans.TC_Currency,
claimstatus.CF_MonthEndStatus,
exposurestatus.CF_MonthEndStatus
```

The aggregation in the Claim DataMart Fact Table works as follows:

1. **For Each Unique Group (Claim, Exposure, Coverage, etc.)**
    - The table aggregates monthly payments, reserves, and recoveries.
    - If any element in the group changes (e.g., exposure, coverage, producer, region), a new row is generated for that month.
2. **How the Monthly Aggregation Works**
    - The query computes financial metrics per month.
    - If there are multiple transactions for a given claim in a month, they are summed to provide a monthly total.
    - The `GROUP BY` ensures that each unique combination of grouping elements gets a separate row.
3. **New Rows When Elements Change**
    - If a claim moves to a different coverage, region, producer, or exposure, it gets a new row for that month.
    - This ensures that changes in claim attributes are captured at a monthly granularity.

The calculation column queries are using window function for ITD, QTD, YTD timeline range calculation. As we are getting the month level aggregation through our group by, we are not using window function for MTD. Let's take example of indemnity payment Quarterly (refer below query), the windowing function SUM will run over the monthly rows for that particular combination of group by, and then do a cumulative sum for QTD.

1. **Partitioning by** `fk_datamartdimkeys_skey`
2. **Grouping by All Key Elements (**`Coverage`**,** `Exposure`**,** `Region`**, etc.)**
3. **Calculating Monthly Aggregates (**`SUM() GROUP BY month.bkey, group_elements`**)**
4. **Generating Separate Rows If Any Group Element Changes**
5. The **window function** (`SUM() OVER()`) in the Claim DataMart Fact Table is used to calculate **cumulative aggregates** for different time periods (**QTD, YTD, ITD**) without collapsing the dataset. Below is the example of QTD calculation for indemnity payment.

```
  sum(sum(coalesce(transline.CF_paymentindemnityamount, 0.0)))
  OVER(PARTITION BY wrk_integration.fk_datamartdimkeys_skey, month.quarterid
order by month.bkey rows unbounded preceding)
  as CF_qtd_paymentindemnityamount,
```

The Claim Fact Table is designed to provide a structured, time-based aggregation of claim transactions, ensuring accurate tracking of payments, reserves, and recoveries across various dimensions. By leveraging window functions and partitioning, it enables efficient MTD, QTD, YTD, and ITD calculations, supporting financial reporting and claims analytics.

Please Reach to Pod Dwarka for any queries, we will be happy to answer.